This demo contains:

- **Generating a One-Way Frequency Report**
- **Creating a Bar Chart**

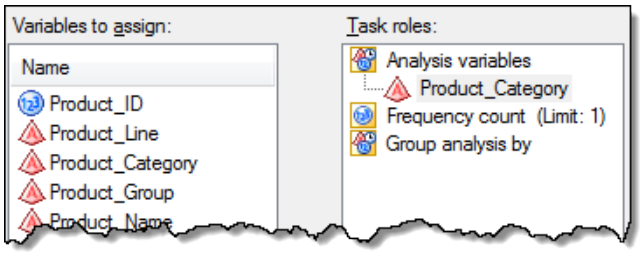
Generating a One-Way Frequency Report

Orion management wants a report that summarizes the number of products offered for each category. Use the One-Way Frequencies task to generate the report.

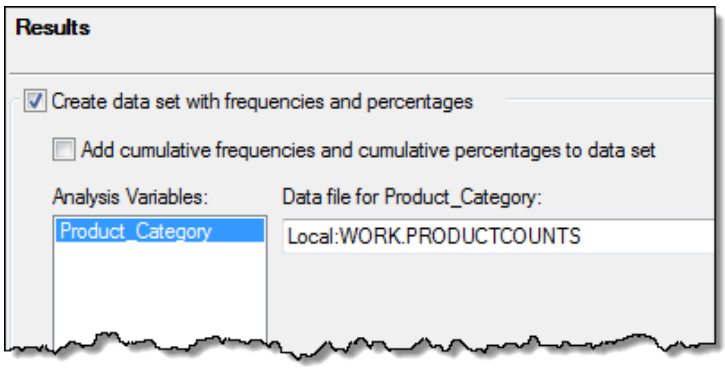
1. Create a new process flow and rename it as **Getting Started with Tasks**.
2. Add the **products** SAS data set from the course data.
3. To open the One-Way Frequencies task, select **Describe** ⇨ **One-Way Frequencies**.

   You can also select **Tasks** ⇨ **Describe** ⇨ **One-Way Frequencies** from the menu bar.

4. Drag **Product_Category** from the Variables to assign pane and drop it onto the **Analysis variables** role in the Task roles pane.



5. Select **Statistics** in the Selection pane. To include only the frequency and percent statistics, select **Frequencies and percentages** in the Frequency table options pane.
6. Select **Results** in the Selection pane. To create a data set including the frequency counts and percentages, select the **Create data set with frequencies and percentages** check box. Click **Browse** and enter **ProductCounts** in the **File name** field. Click **Save**.



7. Select **Titles** in the Selection pane. To modify the title, clear the **Use default text** check box. In the **Text** field, delete the default title of **One-Way Frequencies Results** and type **Number of Products per Category**.

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In the Titles window, when **Analysis** is selected in the Section pane, the Text area displays only your title text. To display the footnote text, select **Footnotes** in the Section pane.

8. To give the One-Way Frequencies task icon a more descriptive name in the project, select **Properties** in the Selection pane and click **Edit**. Enter **Products per Category** in the **Label** field. Click **OK**.

9. Click **Run** to generate the report and examine the results. View the report on the Results tab. Click the **Output Data** tab to view the data set.

   Partial Output

Number of Products per Category

The FREQ Procedure

Product_Category	Frequency	Percent
Assorted Sports Articles	68	14.14
Children Sports	27	5.61
Clothes	108	22.45
Golf	56	11.64
Indoor Sports	15	3.12

10. To remove the title, **The FREQ Procedure**, that is added to the output automatically, select **Tools** ⇨ **Options**. Select **Tasks General** from the Selection pane. Clear the **Include SAS procedure titles in results** check box. Click **OK** to close the Options window.

    The Tasks General pane also includes an option that enables you to change or delete the default footnote.

11. To rerun the task, select **Refresh** on the Results tab.

Number of Products per Category

Product_Category	Frequency	Percent
Assorted Sports Articles	68	14.14
Children Sports	27	5.61
Clothes	108	22.45
Golf	56	11.64
Indoor Sports	15	3.12
Outdoors	87	18.09
Racket Sports	14	2.91
Running - Jogging	25	5.20
Shoes	34	7.07
Swim Sports	7	1.46
Team Sports	23	4.78
Winter Sports	17	3.53

12. Save the project

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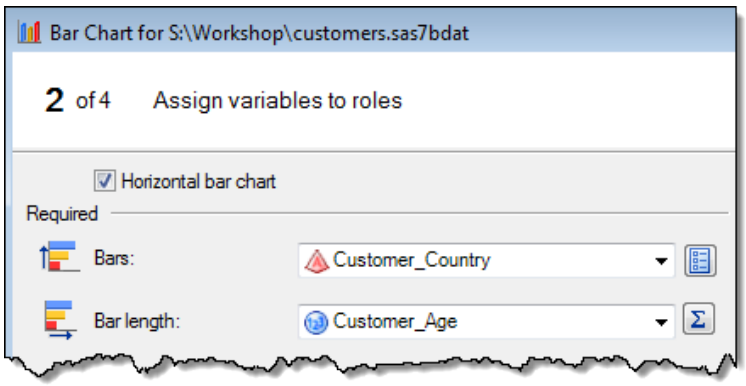
Creating a Bar Chart

To better understand the demographics of the Orion Star customers, management wants to see a comparison of average customer age by country. Use the Bar Chart Wizard and task to create the result.

1. Add the **customers** data set to the process flow.
2. Select **Tasks** ⇨ **Graph** ⇨ **Bar Chart Wizard**.
3. In Step 1, verify that **customers** is the active data set. Click **Next**.
4. In Step 2, select the **Horizontal bar chart** check box. To create a separate bar for each country, select **Customer_Country** in the Bars drop-down list.
5. To order the bars in descending height or length, click . In the Properties window, select **Descending bar height** ⇨ **OK**.
6. To set the length of the bars based on average age, select **Customer_Age** from the Bar length drop-down list.



7. Click , change the statistic to **Average**, and click **OK**. Click **Next**.
8. In Step 3, make the following changes:
   a. Select the **3D chart** check box.
   b. In the Color bars by drop-down list, select **Bar category**.
   c. Select the **Data labels** check box and select **Average** from the drop-down list.
   d. Select the **Use reference lines** check box.
   e. Click **Next**.

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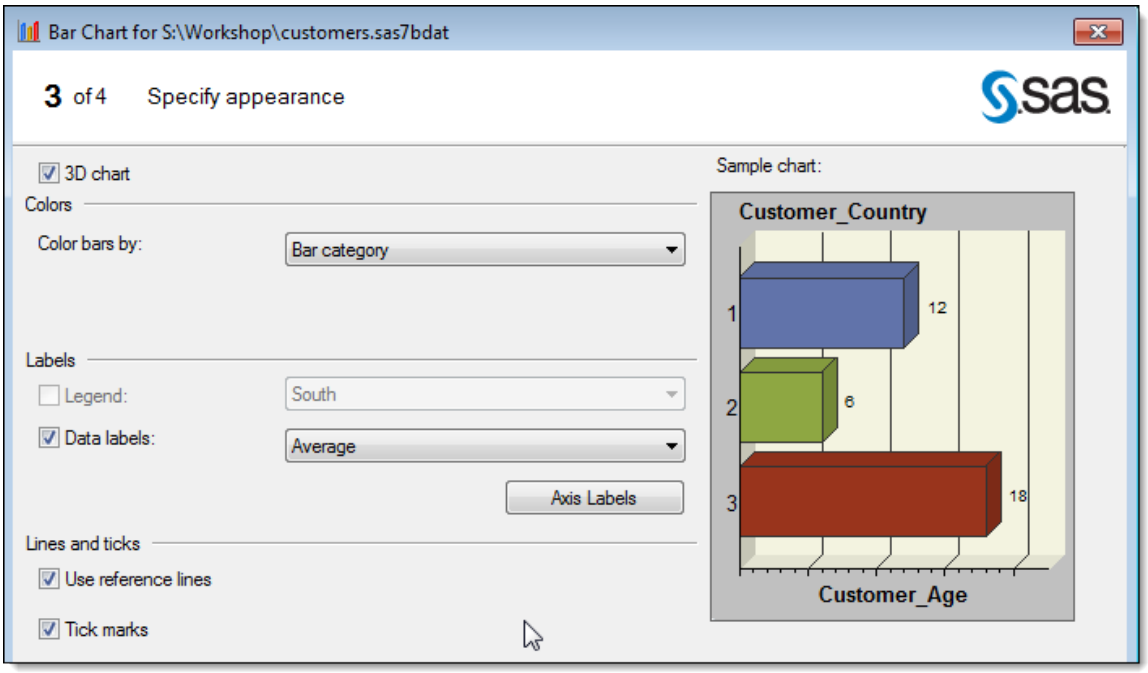
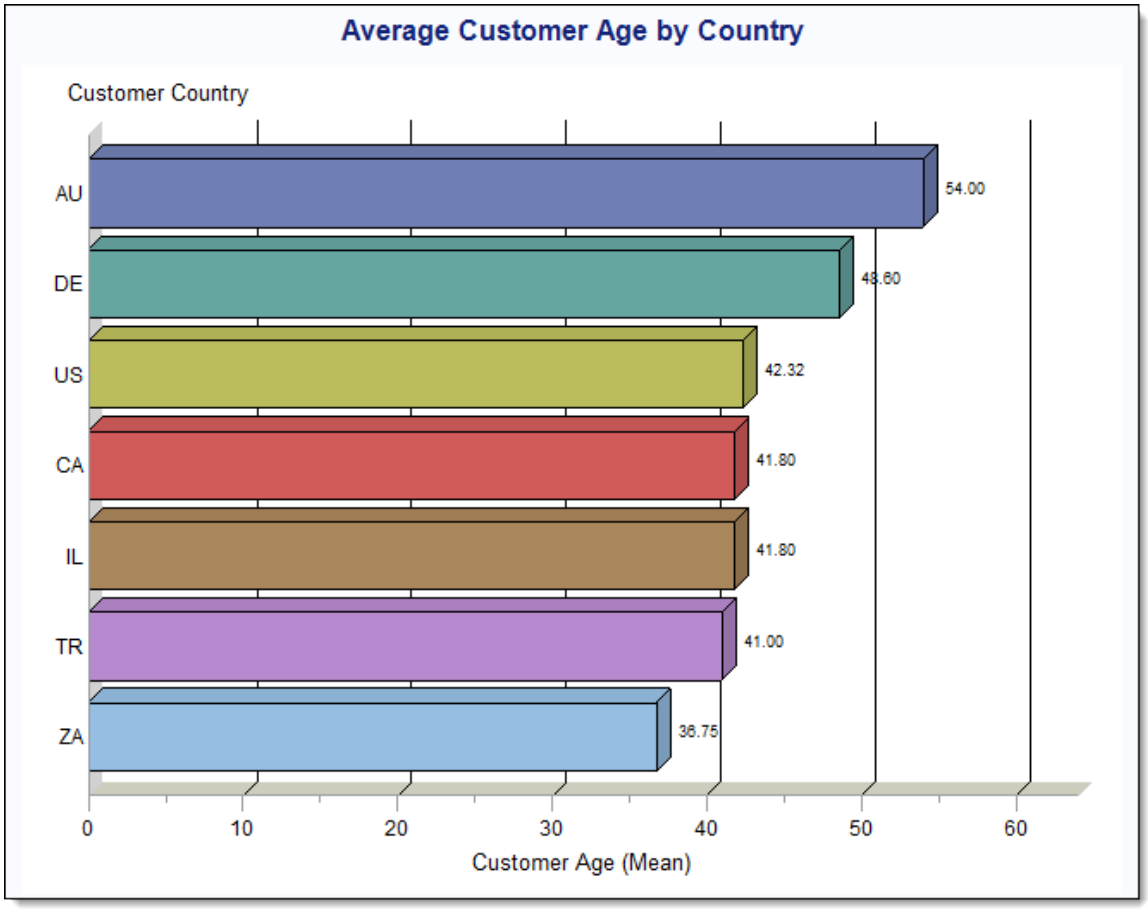


9. In Step 4, change the title to **Average Customer Age by Country** and click **Finish**.



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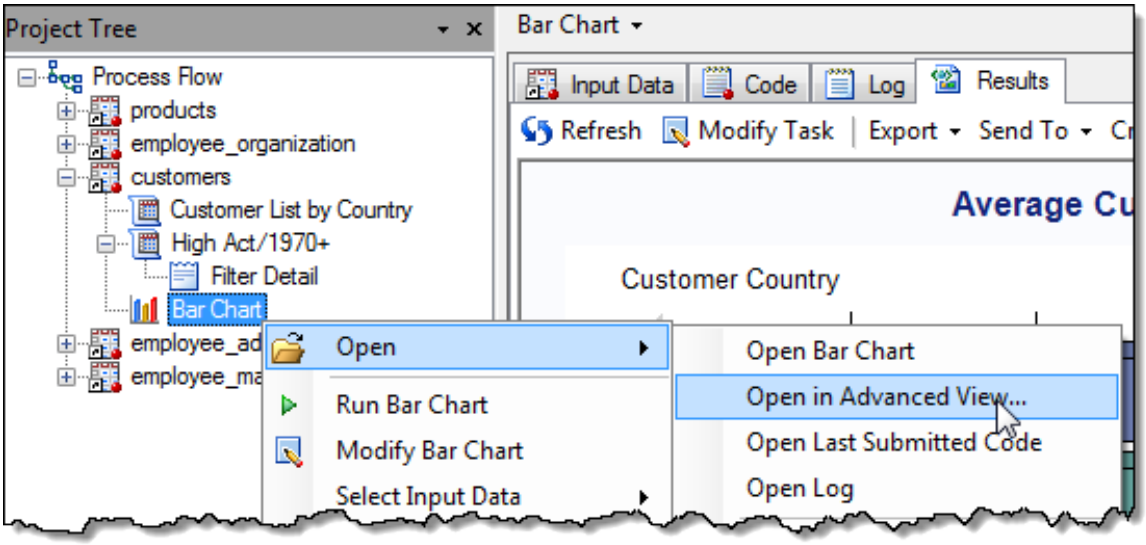
10. To make additional modifications not allowed in the Bar Chart Wizard, open the task in Advanced View. Right-click the **Bar Chart** icon in the project tree or process flow. Select **Open** ⇨ **Open in Advanced View**.

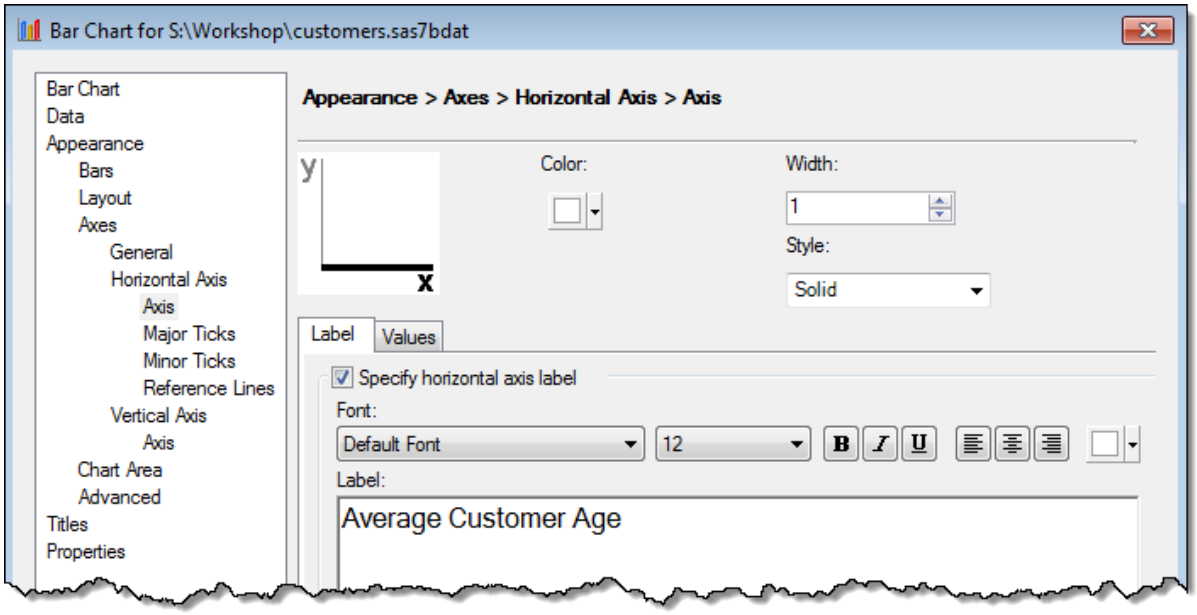


11. In Advanced View, you can apply formats to variables to modify the appearance. Select **Data** in the Selection pane. To format **Customer_Age** so that the data labels are rounded to one decimal place, right-click **Customer_Age** and select **Properties**.

12. Select **Change** to apply a format. In the Formats window, select **Numeric** from the Categories pane and **w.d** from the Formats pane. Change the overall width to **4** and the decimal places to **1**. Click **OK** ⇨ **OK** to return to the task dialog box.

13. Select **Layout** in the Selection pane. Change the shape to **Cylinder**.

14. Select **Horizontal Axis** in the Selection pane. Enter **Average Customer Age** in the Label box and change the font size to **12**.



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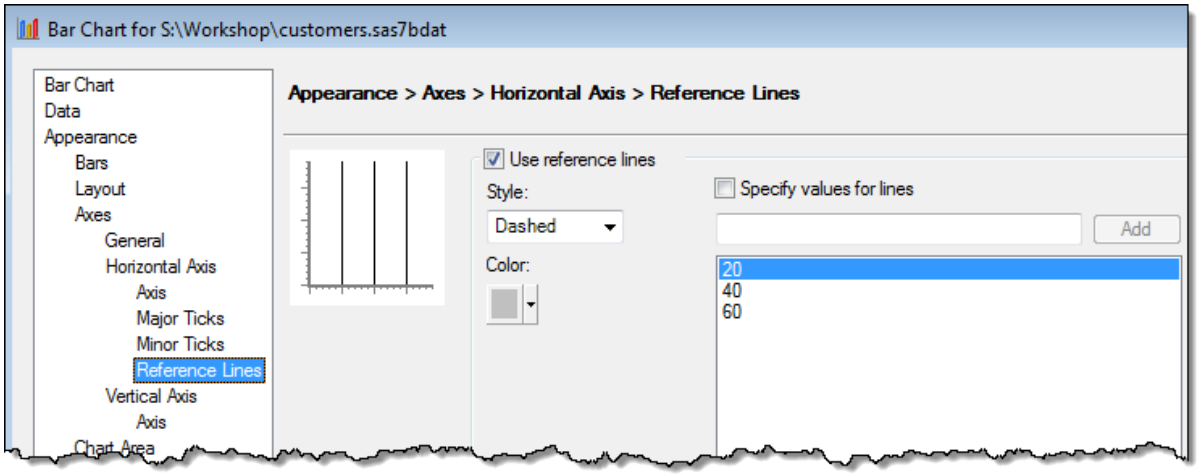
15. Select **Reference Lines** in the Selection pane. Click the **Specify values for lines** check box, enter **20**, and then select **Add**. Repeat for **40** and **60**. Change the style to **Dashed** and the color to light gray.



16. Select **Vertical Axis** in the Selection pane. Enter **Country** in the Label box.

17. Select **Properties** in the Selection pane and select **Edit**. In the General pane, change the task label to **Avg Age/Country**.

18. The default graph format is ActiveX. To be able to interact with the graph after the task runs, it must be in HTML format. In the Results pane, select **Customize result formats, styles, and behavior** and select **HTML**. Click **OK**.

19. Click **Run** to generate the graph and select **Yes** when you are prompted to replace the results. Click the **Results - HTML** tab. To access the ActiveX functionality that enables further exploration and modification of the graph, right-click on the graph and select **Graph Properties**.

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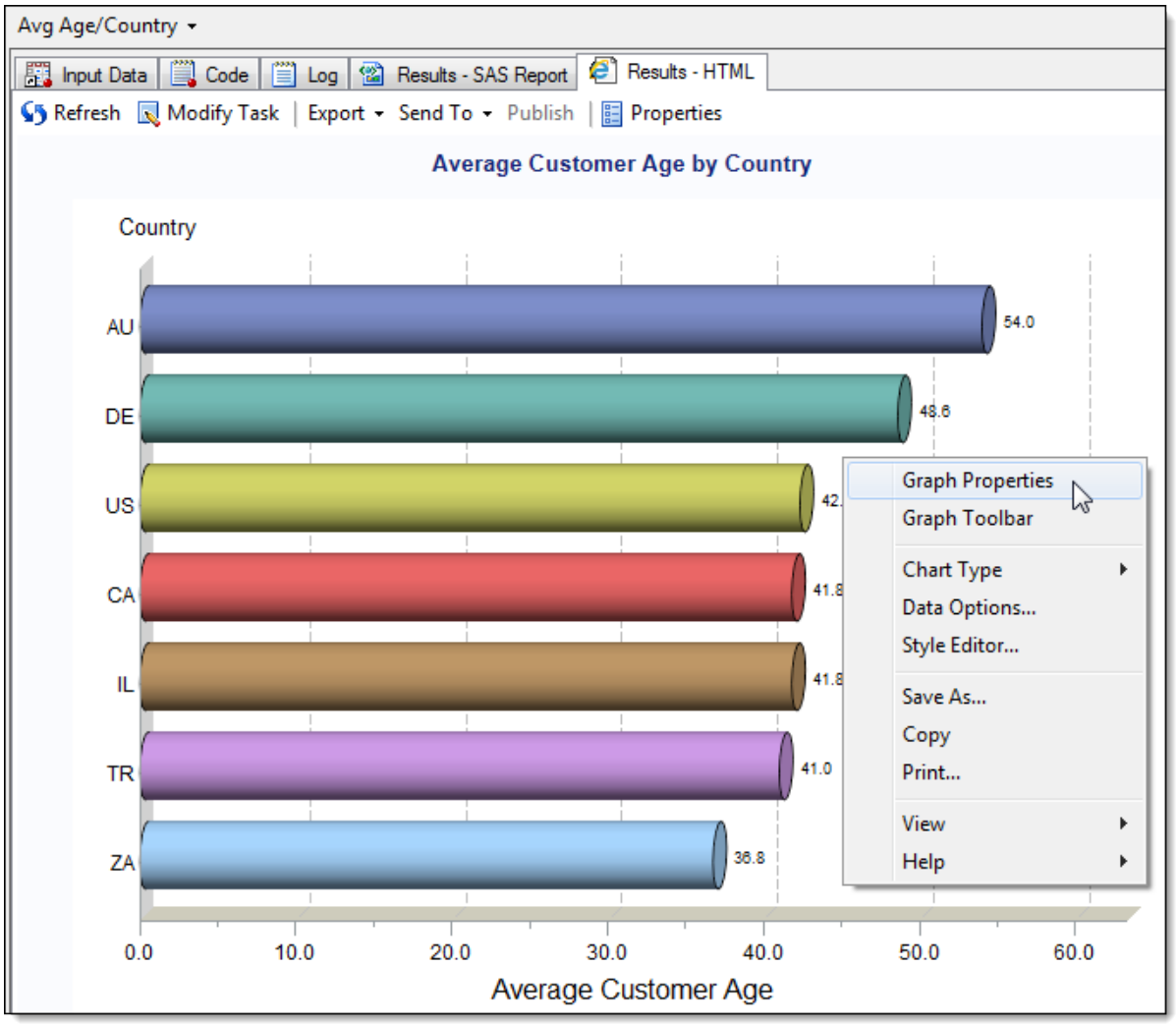

20. On the Graph tab, change the style to **Curve** and a new color scheme is applied.
21. Click the **Bar** tab and select **Font**. Increase the data label font size to **10**. Click **OK** ⇨ **OK** to return to the graph and apply the changes.
22. Another change that can be made is to switch the chart type. Right-click on the graph and select **Chart Type** ⇨ **Vertical Bar**.
23. A toolbar is also available to resize, rotate, and subset the graph. Right-click on the graph and select **Graph Toolbar**. Click and use the mouse to drag the cursor across the top three bars. The chart is redrawn to include only the outlined portion. Click to reset the graph.
24. To save the graph, you can save the image as a JPEG file (saves any interactive changes that you made as a static image) or you can export the graph as an HTML file. (Interactivity is preserved but changes that you made interactively are lost.) To export the HTML result, select **Export** ⇨ **Export HTML - Avg Age/Country**. Navigate to the desired location and select **Save**.

Graphs can also be copied and pasted into Microsoft Word, Excel, and PowerPoint. Right-click on the graph in Enterprise Guide, and select **Copy**. Then right-click in Word, Excel, or PowerPoint, and select **Paste**.

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To maintain ActiveX interactivity in these applications, consider the following:

- Microsoft Excel: The graph automatically retains the ActiveX functionality.
- Microsoft Word: After the graph is pasted into a document, you must exit Design Mode. This can be done in Office 2007 or later by selecting **Developer** ⇨ **Design Mode**.
- Microsoft PowerPoint: To be able to interact with the ActiveX graph while in View mode in PowerPoint, you must first change a graph property before copying and pasting the graph. Right-click on the graph in Enterprise Guide and selecting **Graph Properties**. On the Graph tab, select **Advanced**. In the Advanced Settings window, click the **Display interactive graphs** check box. After you modify the graph properties, copy and paste the graph into PowerPoint.

25. Save the project.

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